

Quantize: Methods and Formulation

The physics implemented in the hybrid quantum–spectroscopic geometry engine

This document records the equations actually implemented in the engine, with a file and line reference for each, together with an honest statement of what has and has not been validated. It is a description of code, not a review of the field: where a quantity is computed by a non-standard route, or carried only as a diagnostic, that is stated rather than smoothed over.

The engine inverts isotopic rotational spectra for molecular geometry. Its distinguishing feature is that the quantum-chemical prior and the spectroscopic data enter a single joint objective, rather than the data being corrected first and fitted afterwards (the semi-experimental route) or ab initio parameters being appended as extra observations (mixed estimation). Section 4 states that difference precisely.

Units are mixed by necessity and are given for every quantity. The engine works internally in MHz for rotational constants, cm^{-1} for vibrational frequencies, Ångstrom for coordinates, amu for masses and Hartree/Bohr² for Hessians; the conversions are stated where they occur.

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1. Vibration–rotation interaction: the $B_e \rightarrow B_0$ correction

`.github/backend/spectral/harmonic_alpha.py` · `rovib_corrections.py` · `rovib_tables.py`

A measured rotational constant belongs to the vibrating molecule in its ground state; the geometry being sought belongs to the hypothetical motionless molecule at the potential minimum. The gap is the vibration–rotation interaction sum, and second-order perturbation theory (VPT2) gives it as three physically distinct contributions, following Mills (1972) and Papoušek & Aliev (1982).

This term is small — measured across this engine's own output it is 0.74% of B — which is why the method works at all. A 25% error in a 0.74% correction is a 0.18% error in B_0 , and since $B \propto 1/r^2$ that is roughly 1 mÅ in a 1.2 Å bond. The accuracy of the whole engine is set here.

Formulation

(1.1) Watson convention

$$B_0^K = B_e^K - \frac{1}{2} \sum_r \alpha_r^K$$

K labels the principal axis (A, B or C) and r the vibrational mode. All constants in MHz. The engine reports the correction as $\delta = (1/2) * \alpha_{\text{sum}}$, so a positive δ raises B .

`harmonic_alpha.py:170`

(1.2) Zero-point amplitude

$$\langle Q_r^2 \rangle_{v=0} = \frac{\hbar}{2\omega_r} \propto \frac{1}{\omega_r}$$

The mean-square normal-coordinate displacement at $v=0$, in amu-weighted Angstrom squared. Every term below carries this $1/\omega$, so a systematically stiff force field biases the whole correction.

`harmonic_alpha.py:313`

(1.3) Term 1 — harmonic (Watson scheme, default)

$$\frac{\partial^2 B_\xi}{\partial Q_r^2} = \frac{3}{2} C \sum_\gamma \frac{(a_r^{\xi\gamma})^2}{I_\xi^2 I_\gamma}$$

$a_r = dI/dQ_r$ is the inertia-tensor derivative. The sum runs over every axis γ , so it needs the OFF-DIAGONAL derivatives that principal-axis diagonalisation discards. Watson's expansion of $\mu = I$ inverse has quadratic coefficient $(3/4) \mu a_\mu a_\mu$ and no second-derivative-of- I term.

`harmonic_alpha.py:362`

(1.4) Harmonic contribution to alpha

$$\alpha_r^{\xi, \text{harm}} = -\frac{\partial^2 B_\xi}{\partial Q_r^2} \langle Q_r^2 \rangle$$

An earlier version of this engine computed a different quantity here. Correcting it moved water's correction error on B from 9852 to 682 MHz and on C from 2402 to 208 MHz.

`harmonic_alpha.py:364`

(1.5) Term 2 — Coriolis

$$\alpha_r^{\xi, \text{cor}} = -\frac{2B_\xi^2}{\omega_r} \sum_{s \neq r} (\zeta_{rs}^\xi)^2 \frac{3\omega_r^2 + \omega_s^2}{\omega_r^2 - \omega_s^2}$$

ζ are the Coriolis coupling constants; the superscript axis is the SAME axis as the constant being corrected. Frequencies in cm⁻¹. This is the only term with a resonance denominator.

`harmonic_alpha.py:439`

(1.6) Term 3 — anharmonic

$$\alpha_r^{\xi, anh} = \langle Q_r^2 \rangle \sum_s \frac{\partial B_\xi}{\partial Q_s} \frac{\phi_{rrs}}{\lambda_s}, \quad \lambda_s = \omega_s^2$$

ϕ_{rrs} are the semi-diagonal cubic force constants, obtained either from the Cartesian cubic tensor or (preferred) by finite difference of the Hessian along each normal mode. Note there is NO Fermi-resonance denominator at first order in alpha.

harmonic_alpha.py:924

(1.7) Semi-diagonal cubic constants

$$\phi_{rrs} = \sum_{ijk} \frac{\partial^3 V}{\partial x_i \partial x_j \partial x_k} D_{ir} D_{jr} D_{ks}$$

D is the mass-weighted normal-mode matrix. These depend only on the electronic surface, not on nuclear masses, so they are computed once and reused across every isotopologue.

harmonic_alpha.py:923

(1.8) Rotational constants from geometry

$$B_K = \frac{h}{8\pi^2 I_K} [\text{MHz}], \quad I_K = \text{eig}_K \left(\sum_a m_a (|r_a|^2 \delta - r_a r_a) \right)$$

The forward model: centre of mass, inertia tensor, principal moments sorted so that $A \geq B \geq C$. Masses in amu, coordinates in Angstrom.

spectral.py:117

(1.9) Expansion-point tracking

$$\alpha(x) \approx \alpha(x_0) + \frac{d\alpha}{ds} \Delta s + \frac{1}{2} \frac{d^2\alpha}{ds^2} \Delta s^2$$

alpha is expanded at the quantum minimum but the joint objective deliberately moves the geometry off it. Three schemes are available (direct, linear, quadratic); 'direct' re-evaluates alpha and won on medians at half the cost.

harmonic_alpha.py:1243

Conventions, units and sign

Frequencies in cm⁻¹, rotational constants in MHz, coordinates in Angstrom, masses in amu, Hessians in Hartree/Bohr squared. The sign convention is Watson's: B₀ is BELOW B_e for a typical molecule, so the correction added to B₀ is positive.

Normal modes come from the mass-weighted Hessian after projecting out rigid-body translation and rotation; the number of rigid modes is 5 for a linear molecule and 6 otherwise, and getting this wrong on a diatomic discards the only real vibration. Modes below a frequency cutoff (default 50 cm⁻¹) are excluded.

Implementation status

COMPLETE for first-order VPT2 with a semi-diagonal cubic force field. All three Mills terms are present and the cubic set is complete for first-order alpha, not truncated.

Frequency scaling (freq_scale) and large-amplitude treatment (lam_freq_cm, which carries a low-frequency mode's entire contribution as uncertainty rather than trusting it) are implemented and optional.

CORIOIS RESONANCES ARE DETECTED BUT NOT TREATED, deliberately. A deperturbation was implemented and then measured: the two terms of a near-degenerate pair carry denominators that are equal and opposite, so they cancel in the sum reaching B₀ however large each is alone. Nudging isocyanic acid's 5.5 cm⁻¹ pair by 1 cm⁻¹ moves alpha_A by 83 MHz out of 6,534,832. The pairs are reported in info['coriolis_resonances']; neither alpha nor sigma is altered.

Validation

Diatomic closed forms: the harmonic term reproduces the exact Dunham result $-6 B_e^2 / \omega_e$ for CO to 1e-4 relative.

Against water's published equilibrium structure the corrected harmonic term reduces the error on B from 9852 to 682 MHz and on C from 2402 to 208 MHz, on both isotopologues.

NOT YET DONE: a term-by-term check against Mills' canonical published form. The expression here was assembled from Mills (1972) and Papousek & Aliev (1982) and validated by its numerical consequences rather than by symbolic comparison.

2. Centrifugal distortion

`.github/backend/spectral/centrifugal_distortion.py · cd_reduction.py`

A rotating molecule stretches, so its energy levels depart from the rigid-rotor pattern. The leading correction is quartic in angular momentum and is governed by the Kivelson–Wilson tau tensor, which this engine builds from the first derivatives of the rotational constants along each normal mode — the same finite-difference path the alpha correction already requires.

A quartic rotational Hamiltonian is over-parameterised: several combinations of tau components produce identical energy levels, so the Hamiltonian must be reparameterised ('reduced') before anything can be fitted. Watson's A and S reductions are the two standard choices.

Formulation

(2.1) Tau tensor

$$\tau_{\alpha\beta\gamma\delta} = -\frac{1}{2} \sum_r \frac{1}{\omega_r^2} \frac{\partial I_{\alpha\beta}}{\partial Q_r} \frac{\partial I_{\gamma\delta}}{\partial Q_r} \times \text{const}$$

Reduction-free, so it can be compared directly against a published tau without worrying which reduction a paper used. Note the 1/omega squared: tau is even more force-field sensitive than alpha.

`centrifugal_distortion.py`

(2.2) Reduction by level matching

$$\min_p \sum_{J,K} (E_{red}(p; J, K) - E_{\tau}(J, K))^2$$

The reduced constants p are obtained by requiring the reduced Hamiltonian to reproduce the energy levels the tau tensor itself generates. This is a linear solve with nothing taken on authority — it replaced a transcribed coefficient table that produced two constants with the wrong sign.

`cd_reduction.py`

(2.3) A-reduction parameters

$$\{D_J, D_{JK}, D_K, \delta_J, \delta_K\}$$

Watson's asymmetric reduction. Its asymmetry operator is $-2 \delta_J P^2 (P_x^2 - P_y^2)$.

`centrifugal_distortion.py:44`

(2.4) S-reduction parameters

$$\{D_J, D_{JK}, D_K, d_1, d_2\}, \quad d_1 = -\delta_J$$

Watson's symmetric reduction, carrying $+d_1 P^2 (P_x^2 + P_y^2)$. The relation $d_1 = -\delta_J$ is the one clean correspondence between the reductions at quartic order, and it is used as a test.

`centrifugal_distortion.py:360`

(2.5) Reduction chosen by asymmetry

$$\kappa = \frac{2B - A - C}{A - C}, \quad |\kappa| \geq 0.9 \Rightarrow S$$

Ray's asymmetry parameter, running from -1 (prolate) to +1 (oblate). The A reduction becomes ill-conditioned as a top approaches symmetry, which is the entire reason S exists.

`centrifugal_distortion.py:419`

Conventions, units and sign

Constants in MHz. Axis ordering $A \geq B \geq C$ throughout, matching the principal-moment sort. Reduction defaults to 'auto' (by kappa) and reduction_order defaults to 4.

Implementation status

Quartic constants: complete, through a derived rather than transcribed reduction.

Order-6 reduction: implemented, NOT the default. It adds sextic-SHAPED terms to the reduction of a purely quartic Hamiltonian; these describe the reduction, not the molecule, and are never written into a correction. Physical sextic constants would require the cubic force field and are not implemented.

CD constants as FIT DATA: implemented and currently INERT. The residual builder skips any isotopologue without an observed cd block, and no reference molecule in the repository carries observed distortion constants, so switching it on is a bit-exact no-op — measured at 8e-9 mA across nine molecules. A warning now says so at construction.

Validation

Against water's experimental A-reduced constants, on an analytic potential: D_J +38.2 against +37.59 (1.6% high), D_K +899 against +973.3 (7.6% low), D_JK -235 against -172.9 (36% low). Signs correct on all five constants.

Ozone's reduction-free tau is reproduced across six isotopologues: 2.3% on tau_aaaa, 9.9% on tau_bbbb, 12.6% on tau_cccc, systematic and isotope-independent to 0.1% — the signature of force-field error rather than an implementation error.

TWO MOLECULES. That is not a validation set.

MEASURED AND REJECTED: choosing the reduction by level-fit residual. S fits water's levels more than twice as well as A and agrees with experiment WORSE, 18.2% against 15.1%. Separately, order 6 cuts the reduction misfit by a third and does not change the agreement with experiment at all (15.1% either way). A better level fit is not a better constant.

3. The small corrections: electronic, Born–Oppenheimer breakdown, mass dependence

`.github/backend/spectral/electronic_g.py · correction_models.py · mass_dependent.py`

Beyond vibration, two further effects separate a measured rotational constant from the one a rigid nuclear framework would give. The electrons do not follow the nuclear rotation perfectly, which shifts B by a term proportional to the rotational g-tensor; and the Born–Oppenheimer separation is itself approximate, which shifts B in a mass-dependent way. Both are small — of order 1 MHz against correction uncertainties of hundreds — but both are isotope-dependent, and isotopic differences are exactly where the structural information lives.

Formulation

(3.1) Full semi-experimental constant

$$B_e^{SE} = B_0 + \Delta B_{vib} - \Delta B_{elec} - \Delta B_{BOB}$$

The assembly the engine performs per isotopologue per axis. All terms in MHz.

`correction_models.py:307`

(3.2) Electronic correction

$$\Delta B_{elec}^\alpha = -\frac{m_e}{m_p} g_{\alpha\alpha} B_{obs}$$

g is the rotational g-tensor component for the same principal axis, dimensionless in nuclear magneton units, computed by coupled-perturbed Hartree-Fock. The electron-to-proton mass ratio (1/1836) is what makes this small.

`correction_models.py:312`

(3.3) Born–Oppenheimer breakdown

$$\Delta B_{BOB} = -B_{obs} \sum_a \frac{m_e}{M_a} u_a$$

The sum runs over atoms; M_a is the atomic mass and u_a an element-specific BOB parameter (hydrogen carries the largest at about 0.015, oxygen about 0.002).

correction_models.py:373

(3.4) Asymmetry of the g-tensor across isotopologues

$$\delta(\Delta B_{elec}) \propto g_{\alpha\alpha}(B_{obs}^{(i)} - B_{obs}^{(j)})$$

The g-tensor is a molecular property and is essentially isotope-independent, so the correction scales with each isotopologue's own B. This is why it cannot simply be absorbed into a constant offset.

electronic_g.py

Conventions, units and sign

All in MHz. Signs follow the assembly equation above: both corrections are SUBTRACTED when building B_e from B_0 , so the helper functions return a signed delta to ADD.

For a linear molecule the component along the molecular axis is identically zero and only $g_{\perp}$ exists; the A entry is a 0/0 artifact of principal-axis diagonalisation and must not be used.

Implementation status

Electronic: implemented, optional, off by default. The g-tensor comes from pyscf's CPHF implementation; a per-isotopologue path exists.

BOB: implemented with built-in u-parameters including heavy atoms. Optional, off by default.

Mass-dependent (Watson r_m -type) structures: implemented as an alternative structural model.

Validation

BOB, MEASURED AND NEGATIVE: sweeping the hydrogen u-parameter from -0.015 to +0.045 across four molecules changes the fitted structure by 0.00%. Across all nine benchmark molecules in both prior configurations the hybrid moved by 0.000 mA on every one. Only acetyl fluoride has a genuine interior minimum, and it sits exactly at the built-in 0.015. The correction is right to have, since one that silently returns zero is a liability, but it is not a lever on accuracy until correction sigmas fall to single-digit MHz.

Electronic: the sign convention is checked against OCS ($g_{\perp} = -0.028711$) and water (D2O $G_{bb} = +0.36009$ measured against +0.341 computed, right sign, 5% low, which is expected Hartree-Fock behaviour). A multi-molecule validation harness has NOT been built.

The literature note in dev/g_tensor_references.py is worth reading before trusting any published g value: four confirmed text corruptions were found in deposited abstracts during that search.

4. The joint objective

.github/backend/spectral/spectral.py · spectral_model.py · SVD.py · quantize.py · geometric_bridge.py

This is the research contribution. The standard semi-experimental route is sequential: correct every measured constant to B_e first, then fit a geometry to the corrected data. Demaison's mixed estimation instead appends ab initio structural parameters to the fit as additional observations with assigned uncertainties. This engine does neither. It places the quantum chemistry and the spectroscopy in one objective function and optimises the geometry against both simultaneously, so the correction's expansion point can follow the geometry as it moves and the quantum term acts as a physical restoring force rather than a set of pseudo-data.

The practical consequence is that the quantum prior is a potential-energy surface, not a list of target bond lengths. It is informative everywhere, including in the directions the rotational constants cannot see.

Formulation

(4.1) Total objective

$$\chi^2(x) = \sum_{i,K} \frac{(B_K^{calc}(x) - B_K^{e,SE})^2}{\sigma_{i,K}^2} + \chi_{prior}^2(x) + w_{cd} \sum \frac{(D^{calc} - D^{obs})^2}{\sigma_D^2}$$

i runs over isotopologues, K over the axes each one supplies. x is the Cartesian geometry. The third block is present but inert for want of observed distortion constants.

spectral.py:960

(4.2) Quantum prior as a harmonic surface

$$\chi_{prior}^2(x) = \frac{1}{\sigma_x^2} [g^T(x - x_0) + \frac{1}{2}(x - x_0)^T H(x - x_0)]$$

g and H are the gradient and Hessian from the quantum backend. sigma_x is the prior width in Angstrom, set per level of theory or per molecule from the offset residuals.

quantize.py

(4.3) Prior re-centring

$$x_0 \rightarrow x_{target}$$

Without re-centring, the quantum term's minimum stays at the level of theory's own minimum, so a bias-corrected prior is discarded on the first step. Measured on vinyl fluoride: 18.01 mA with the centre left alone against 7.95 mA with it moved.

accuracy_upgrades_benchmark.py:374

(4.4) Weighted normal equations

$$(J^T J) \Delta x = -J^T r, \quad J_w = \frac{1}{\sigma} \frac{\partial B}{\partial x}$$

Gauss-Newton on the weighted residual. Robust weights and per-component weights multiply both J and r so the normal equations stay consistent.

spectral.py:966

(4.5) Rigid-mode projection

$$P = 1 - \sum_m v_m v_m^T, \quad \Delta x \rightarrow P \Delta x$$

Translations and rotations leave the energy unchanged and must be removed. The projector is built UNWEIGHTED, because the gradient and Hessian it acts on are plain Cartesian derivatives; weighting by the square root of the mass would project out the wrong subspace.

quantize.py

(4.6) Chi-squared rescaling

$$\sigma \rightarrow \sigma \sqrt{\chi_v^2} \quad \text{if } \chi_v^2 > 1$$

If the fit cannot reproduce the data within the stated uncertainties, the uncertainties were understated; the rescaling makes the reported precision honest rather than the fit look good. Applied iteratively, capped at three passes.

quantize.py

Conventions, units and sign

Residuals are formed in MHz and divided by sigma in MHz, so chi-squared is dimensionless. The prior term is formed in Angstrom and divided by sigma_x in Angstrom. A relative floor is applied to every observation sigma as a fraction of the constant, so that an implausibly tight quoted uncertainty cannot dominate the fit.

Implementation status

Complete and the default path. The engine carries its own optimiser; an alternative route through geomeTRIC with the spectroscopic restraint built in is also implemented, for internal coordinates and better convergence on floppy systems.

Conditioning is monitored by singular-value decomposition, and an identifiability analysis reports parameters the data cannot determine.

Validation

Benchmarked against mixed estimation on nine molecules at two levels of theory. At RHF/6-31G the hybrid is best of the three legs on 6 of 9 and beats mixed estimation on 8 of 9; at B3LYP/6-31G(d), 5 of 9 and 7 of 9. The margin improves with the level of theory, which is the property that matters — it is not merely rescuing a bad prior.

CAVEAT ON ALL OF THE ABOVE: every benchmark ran at a level of theory no one would publish a structure at, and eight of the nine reference structures are 1950s-60s substitution structures whose own Costain floor is about 4 mA — comparable to the errors being measured. On fluoroacetylene, scoring the identical fitted geometry against a published equilibrium structure instead of the substitution one takes the error from 4.44 to 2.41 mA with nothing in the engine changed.

5. Priors, bias correction and identifiability

`.github/backend/spectral/bond_offsets.py · internal/ · priors/`

A Gaussian prior centred on the level of theory's own geometry asserts that the theory is unbiased — noisy, but not systematically wrong. That assertion is false: a given method and basis set gets a C–F bond wrong in a reproducible direction. The bond-class offset machinery measures that bias per bond class on a calibration set and moves the prior's centre accordingly, then narrows the prior's width to match the bias that has been removed.

Because the offsets are measured on the same molecules the engine is scored against, every molecule is corrected using offsets pooled from the OTHER molecules only. A bond class with no other member is left uncorrected. This makes the reported gain a lower bound rather than an artefact.

Formulation

(5.1) Bond-class offset

$$\delta_c = \text{median}_{m \in M} (r_{m,c}^{\text{ref}} - r_{m,c}^{\text{theory}})$$

Measured per bond class c over the calibration molecules. Applied by displacing the prior's centre along that internal coordinate. In Angstrom.

`bond_offsets.py`

(5.2) Leave-one-out protocol

$$\delta_c^{(m)} = \text{median}_{m' \neq m} (r_{m',c}^{\text{ref}} - r_{m',c}^{\text{theory}})$$

The offset applied to molecule m never sees molecule m . Without this the benchmark would be scoring the engine on data used to tune it.

`bond_offsets.py`

(5.3) Prior width follows the correction

$$\sigma_x \rightarrow \sigma_x^{(m,c)} = \text{spread}(\{r^{\text{ref}} - r^{\text{theory}}\}_c)$$

Bond by bond, because a class the leave-one-out could not calibrate is not narrowed at all. Quoting the corrected spread for an uncalibrated class claims a precision the prior does not have and cost chlorofluoromethane 13.16 to 19.71 mA when tried.

`bond_offsets.py`

(5.4) Identifiability from the Jacobian

$$\text{cond}(J) = \frac{s_{\max}}{s_{\min}}, \quad J = USV^T$$

Singular values of the weighted Jacobian. A direction with a vanishing singular value is a parameter combination the data cannot determine; the right response is a prior on it, not a tighter convergence criterion.

`internal/identifiability.py`

Conventions, units and sign

Lengths in Angstrom throughout. Offsets are signed as reference minus theory, so a positive offset lengthens the prior's bond.

Implementation status

Complete. Bond-class offsets, leave-one-out pooling, per-molecule prior widths, internal-coordinate priors with a small library of defaults, several prior-weighting schemes and a Bayesian width-tuning route are all implemented. Offsets are the single largest measured accuracy gain in the engine.

Validation

At RHF/6-31G, switching offsets on moves the nine-molecule mean from 11.44 to 8.59 mÅ and the median from 13.39 to 7.22 mÅ, under leave-one-out. Individual molecules: vinyl fluoride 18.33 to 6.73, fluoroacetylene 13.59 to 4.42.

It is not uniformly good — chlorofluoromethane goes the wrong way, 13.39 to 20.15, and it is the molecule with the least data per parameter in the set.

6. Large-amplitude motion and internal rotation

`.github/backend/torsion/` (17 modules)

VPT2 expands in the vibrational coordinate about a single minimum. A methyl torsion at 100–250 cm⁻¹ has a zero-point amplitude far outside the range where that expansion holds, and the correct statement about such a mode's contribution to alpha is not a number but an interval. The engine offers two responses: an honest one that is cheap, and a physical one that is not.

The cheap response treats any mode below a threshold as large-amplitude and moves its entire contribution from the value into the uncertainty. The physical response solves the torsional problem properly and averages the rotational constants over the torsional wavefunction.

Formulation

(6.1) Large-amplitude mode carried as uncertainty

$$\omega_r < \omega_{LAM} \Rightarrow \alpha_r \rightarrow 0, \quad \sigma^2 \rightarrow \sigma^2 + \alpha_r^2$$

The perturbative value is discarded and its magnitude becomes a width. Default threshold is 0 (disabled); a typical setting for a methyl rotor is 250 cm⁻¹.

`harmonic_alpha.py:635`

(6.2) Torsional Hamiltonian

$$\hat{H}_{tors} = F\hat{p}_\alpha^2 + \frac{V_3}{2}(1 - \cos 3\alpha) + \frac{V_6}{2}(1 - \cos 6\alpha)$$

F is the reduced internal-rotation constant, alpha the torsional angle. Solved in a free-rotor basis; V3 and V6 come from a relaxed torsional scan on the quantum surface.

`torsion/torsion_hamiltonian.py`

(6.3) Torsional averaging of a rotational constant

$$\langle B_K \rangle_{v_t} = \int \psi_{v_t}^*(\alpha) B_K(\alpha) \psi_{v_t}(\alpha) d\alpha$$

The expectation value over the torsional wavefunction replaces the rigid value, and the difference feeds back into the correction table in place of the perturbative alpha for that mode.

torsion/torsion_average.py

(6.4) Reduced-axis torsion–rotation coupling

$$\hat{H} = \hat{H}_{rot} + \hat{H}_{tors} - 2F\rho\hat{p}_a\hat{J}_a + F\rho^2\hat{J}_a^2$$

The RAM-lite scaffold. rho measures the projection of the internal-rotor angular momentum on the a axis. 'Lite' denotes that the full Hamiltonian is truncated.

torsion/torsion_rot_hamiltonian.py

Conventions, units and sign

Torsional angles in radians internally, degrees at the interface. V3, V6 and F in cm⁻¹. Torsional symmetry is handled by the appropriate periodicity and symmetry number.

Implementation status

MIXED, and this should be read carefully. The sigma-widening route is complete and validated. The 1D torsional Hamiltonian, relaxed-scan fitting, torsional symmetry handling, 1D and 2D averaging, intensities and a torsional uncertainty treatment are implemented. The reduced-axis torsion–rotation Hamiltonian is described in its own source as a SCAFFOLD, not a finished treatment.

None of the benchmark molecules in the standard set requires the full machinery; acetyl fluoride and fluoroethane have methyl rotors and are the natural test cases.

Validation

Tested on acetaldehyde and a dedicated large-amplitude benchmark set. The sigma-widening route behaves as designed: it widens the molecules with rotors and leaves the others alone.

The full torsion-rotation treatment has NOT been validated against experimental torsional splittings across a molecule set.

7. Diagnostics and uncertainty quantification

.github/backend/kraitchman.py · symmetry.py · uncertainty/ · uncertainty_v2/

Three diagnostics decide whether a fitted structure should be believed. Kraitchman's substitution equations give an independent, model-free estimate of each atom's position from isotopic differences alone. The inertial defect measures how much vibrational contamination survives in a planar molecule's corrected constants, and so acts as a direct probe of residual correction bias. And the uncertainty machinery propagates the observation uncertainties through to the structure, analytically and by resampling.

Formulation

(7.1) Kraitchman substitution coordinate

$$|a| = \sqrt{\frac{\Delta P_a}{\mu}}, \quad \mu = \frac{M\Delta m}{M + \Delta m}$$

Delta P_a is the change in the planar moment on substitution and mu the reduced mass of the substitution. The square root loses the sign and becomes imaginary for an atom near a principal plane — the classic failure mode.

kraitchman.py

(7.2) Inertial defect

$$\Delta = I_c - I_a - I_b$$

Zero for a rigid planar molecule. In amu Angstrom squared. A non-zero value measures out-of-plane vibrational contribution, so after a correct vibrational correction it should approach zero.

`kraitchman.py:62`

(7.3) Defect ratio as a bias probe

$$R = \frac{|\Delta_{corrected}|}{|\Delta_{observed}|}$$

The RATIO carries the signal, not the absolute residual. An earlier version used the absolute residual as a sigma floor and damaged the molecules whose corrections work.

`rovib_corrections.py`

(7.4) Analytic parameter covariance

$$\text{Cov}(x) = (J^T W J)^{-1}, \quad W = \text{diag}(1/\sigma^2)$$

Standard weighted least-squares propagation, optionally rescaled by the reduced chi-squared. Internal-coordinate uncertainties follow by the chain rule through dq/dx.

`uncertainty/core.py`

Conventions, units and sign

Moments in amu Angstrom squared, coordinates in Angstrom. Symmetry constraints are imposed by projecting the Cartesian step onto the symmetric subspace of the chosen point group.

Implementation status

Kraitchman, the inertial defect and the analytic covariance route are complete and default. A second-generation uncertainty package adds bootstrap resampling, MCMC posterior sampling, posterior summaries and convergence diagnostics; it is optional.

Point-group symmetry constraints, model selection and multistart global search are implemented and optional.

Validation

The inertial-defect probe was measured rather than assumed: used as a sigma SCALE it moved vinyl fluoride 7.95 to 7.21 mA and fluoroethane 8.30 to 7.40, left acetyl fluoride essentially unchanged, and did not touch the other six molecules at all. That selectivity is the property that matters, because the earlier absolute-floor version bought a similar headline gain by damaging the molecules whose corrections work.

Kraitchman is checked against published substitution structures.

8. Summary of status

What follows is the honest ledger. The physics of the correction chain is standard and complete to first order in VPT2. The evidence base is not yet at publication standard, for two reasons that are both fixable and neither of which is a physics problem.

FIRST, every benchmark in this repository ran at a level of theory — RHF/6-31G, or at best B3LYP/6-31G(d) — at which no one would publish a structure. The engine's own default is CCSD(T)/cc-pVTZ and it has never been benchmarked there. Published analyses of the semi-experimental method put a CCSD(T)/cc-pVTZ cubic force field at roughly 0.09% error in the correction sum against 0.21% for MP2/cc-pVTZ; this engine's claimed correction uncertainty corresponds to about 0.17%. Running at the default level is the single largest available gain and requires no new code.

SECOND, eight of the nine reference structures are substitution or effective structures from the 1950s and 60s, whose own uncertainty is of order 4 mA — comparable to the errors being measured. Only water has a true

equilibrium reference, and water is the only sub-milliångstrom result in the set. On fluoroacetylene, scoring the identical fitted geometry against a published equilibrium structure rather than the substitution one takes the error from 4.44 to 2.41 mÅ.

Several negative results are recorded in the code and should not be re-derived: Coriolis resonance deperturbation, Born-Oppenheimer u-parameters, order-6 reduction, and reduction choice by level-fit residual were each implemented, measured, and found not to help. Distortion constants as fit data are implemented and inert for want of observed data.

Formulation

(8.1) Error leverage, the reason this works at all

$$\frac{\delta r}{r} \approx \frac{1}{4} \frac{\sum \delta \alpha_r}{B_0}$$

The correction's error is diluted by B_0 rather than by the correction. Since the correction is about 0.74% of B in this engine's own output, a 25% error in it gives about 1 mÅ in a 1.2 Å bond. Reaching 0.1 mÅ requires the correction to about 2%, which is a force-field problem, not a spectroscopy or a fitting problem.

measured across 241 correction rows

Implementation status

COMPLETE AND DEFAULT: vibration-rotation correction (all three Mills terms), quartic centrifugal distortion with derived A and S reductions, the joint objective, bond-class offsets with leave-one-out, Kraitchman, inertial-defect diagnostics, analytic uncertainty.

COMPLETE AND OPTIONAL: electronic g-tensor correction, BOB, mass-dependent structures, frequency scaling, large-amplitude sigma widening, expansion-point tracking, symmetry constraints, multistart, bootstrap and MCMC uncertainty, geomeTRIC route.

PARTIAL OR SCAFFOLD: reduced-axis torsion-rotation Hamiltonian.

IMPLEMENTED BUT INERT: distortion constants as fit data (no observed data in the repository).

NOT IMPLEMENTED: physical sextic distortion constants (requires the cubic force field).

Validation

VALIDATED: harmonic alpha against diatomic closed forms and water's equilibrium structure; CD reduction against water's experimental constants and ozone's reduction-free tau; the joint objective against mixed estimation on nine molecules at two levels; bond offsets under leave-one-out; the inertial-defect probe by its selectivity.

NOT VALIDATED: the alpha expression against Mills' canonical published form, symbolically; the g-tensor across more than a sign check on two molecules; distortion constants beyond two molecules; the full torsion-rotation Hamiltonian against measured splittings.

The ranked list of things that would most improve the engine: (1) benchmark at the default CCSD(T)/cc-pVTZ rather than RHF/6-31G; (2) replace the substitution reference structures with published equilibrium ones; (3) obtain observed distortion constants so the fit rows do something; (4) check alpha against Mills symbolically.